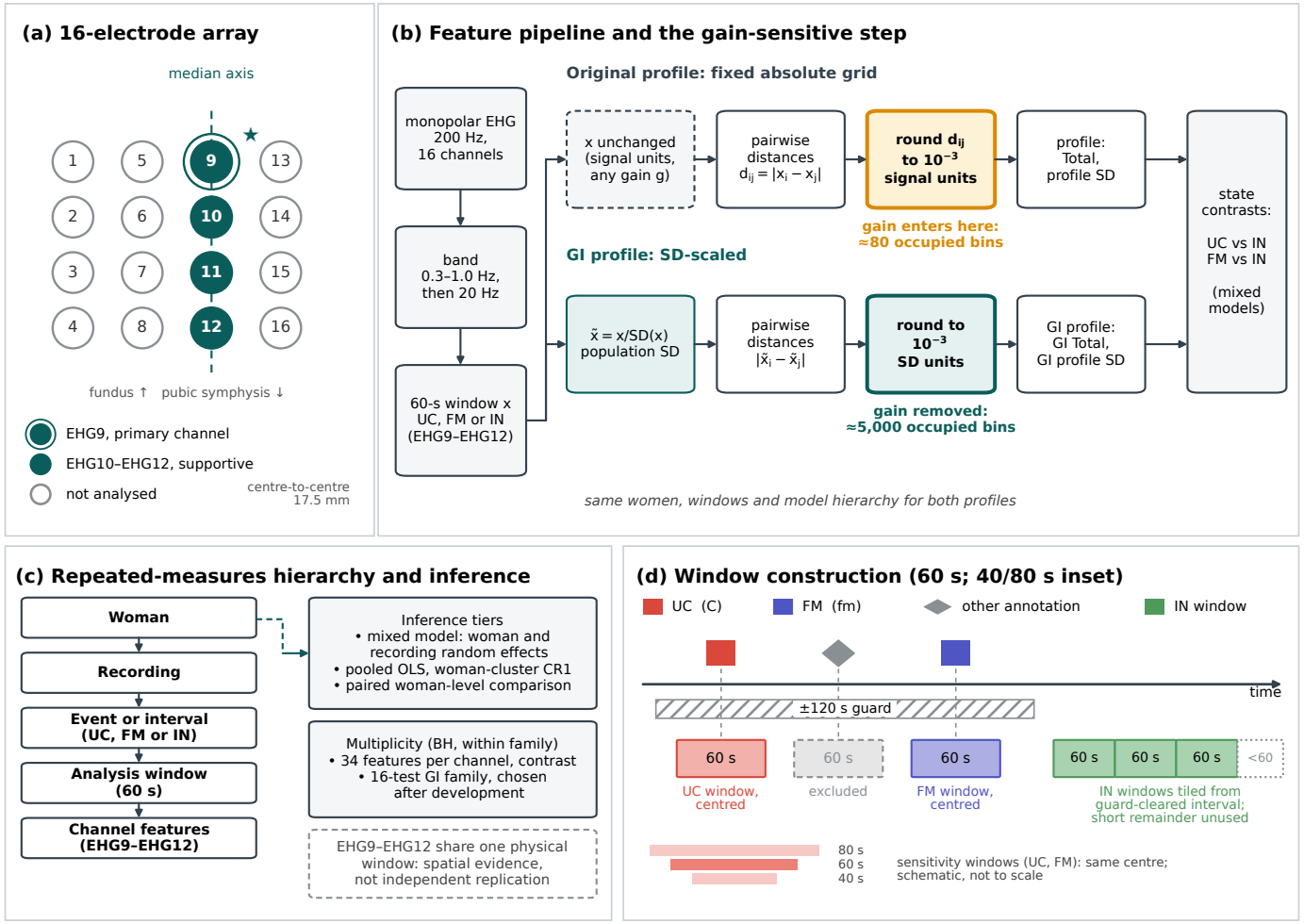
(a) 16-electrode array
median axis
1
5
9
13
2
6
10
14
3
7
11
15
4
8
12
16
fundus ↑
pubic symphysis ↓
EHG9, primary channel
EHG10–EHG12, supportive
not analysed
centre-to-centre
17.5 mm
(b) Feature pipeline and the gain-sensitive step
Original profile: fixed absolute grid
monopolar EHG
200 Hz,
16 channels
band
0.3–1.0 Hz,
then 20 Hz
60-s window x
UC, FM or IN
(EHG9–EHG12)
x unchanged
(signal units,
any gain g)
pairwise
distances
$d_{ij} = |x_i - x_j|$
round d_{ij}
to 10^{-3}
signal units
profile:
Total,
profile SD
gain enters here:
≈80 occupied bins
GI profile: SD-scaled
$\tilde{x} = x/SD(x)$
population SD
pairwise
distances
$|\tilde{x}_i - \tilde{x}_j|$
round to
10^{-3}
SD units
GI profile:
GI Total,
GI profile SD
gain removed:
≈5,000 occupied bins
state
contrasts:
UC vs IN
FM vs IN
(mixed
models)
same women, windows and model hierarchy for both profiles
(c) Repeated-measures hierarchy and inference
Woman
Recording
Event or interval
(UC, FM or IN)
Analysis window
(60 s)
Channel features
(EHG9–EHG12)
Inference tiers
• mixed model: woman and
recording random effects
• pooled OLS, woman-cluster CR1
• paired woman-level comparison
Multiplicity (BH, within family)
• 34 features per channel, contrast
• 16-test GI family, chosen
after development
EHG9–EHG12 share one physical
window: spatial evidence,
not independent replication
(d) Window construction (60 s; 40/80 s inset)
UC (C)
FM (fm)
other annotation
IN window
time
±120 s guard
60 s
60 s
60 s
60 s
60 s
60 s
<60
UC window,
centred
excluded
FM window,
centred
IN windows tiled from
guard-cleared interval;
short remainder unused
80 s
60 s
40 s
sensitivity windows (UC, FM): same centre;
schematic, not to scale